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# Steering Wheel Torque Rendering: Measure of Driver Discrimination Capabilities

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## Abstract

By the action on the steering wheel, the driver has the capability to control the trajectory of its vehicle. Nevertheless, the steering wheel has also the role of information provider to the driver. In particular, the torque level at the steering wheel informs the driver about the interaction between the vehicle and the road. This information flow is natural due to the mechanical chain between the road and the steering wheel. Many studies have shown that steering wheel torque feedback is crucial to ensure the control of the vehicle. In the context of uncoupled steering (steer-by-wire vehicle or driving simulators), the torque rendering on the steering wheel is a major challenge. In addition, of the trajectory control, the quality of this torque is a key for the immersion of drivers in virtual environment such as in driving simulators.

The torque-rendering loop is composed of different steps. At first, a vehicle dynamics model computes the torque level at the steering wheel regarding the vehicle state (steering wheel position, vehicle speed, etc.). The second step is a control strategy, which ensures that the torque-rendering device achieves the torque demand. In this paper, we propose to study the capability of driver to discriminate different torque levels in steady state condition. A specific control strategy was implemented in order to ensure high quality torque rendering. We measured the Just Noticeable Differences for different torque levels: 1, 3 and 5 Nm, in both directions (clockwise and counter-clockwise). This experiment involved 18 subjects. The results of this experiment are compared with the literature and give cues to validate torque rendering control strategies from a driver point of view. It also helps to understand how driver perceives the steering wheel torque.

## Introduction

The primary role of the steering wheel is to allow the driver to manage the trajectory of its vehicle. A second one is to provide feedback to the driver about the environment. This is due to the mechanical link from the steering wheel to the tyre and the road. In fact, a sudden lateral wind or a variation of the road adherence can result in a modification of the vehicle dynamics, which could have effects on the steering wheel behavior.

*Power steering technology.* The concept of power steering appeared in the beginning of the twentieth century. However, car manufacturers introduced such system in the 1950s. Chrysler proposed the “hydraguide”, a hydraulic power steering system in 1951 on the Chrysler Imperial. Such systems assist drivers to turn face mechanical effort needed to handle the steering wheel and then the vehicle trajectory. These systems also allow information filtering sent to the steering wheel like some road vibrations in order to improve the steering feel.

*Advanced Driving Assistance Systems.* In addition, some ADAS such as Lane Departure Warning can have the capability to provide additional information (like steering wheel vibrations) about the vehicle path or the ambient traffic to the driver. Lane Keeping Assistance systems are also able to act on the vehicle steering to avoid critical situations. Besides, a failure of a component of these systems can result on effects on the steering wheel.

*Uncoupled steering.* In the context of Steer-by-Wire vehicle [1, 2] or driving simulators, the steering wheel is mechanically uncoupled and the rendered torque can be easily modified. Then it appears that many factors can modify or deter the torque feedback. The question of the affect on driving performance arises.

*Importance of torque feedback*. In [3], it was observed that during driving in a curve with a constant radius, there was no difference in the steering wheel angle variation whether there was torque rendering or not. In addition no difference was measured on straight roads [3-4]. Both studies were performed on a static driving simulator. But it was also noticed that the presence of torque feedback improves the driving performance after recovering from a skid [3]. In [5] it is proposed that

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increasing the level of tactile feedback on the steering wheel can improves the stability of the vehicle during specific maneuvers.

In [4], subjects had more difficulties to manage a sloped exit ramp without steering wheel torque. In another experiment [6], it appeared the effect of a null torque rendering altered the driving performance.

It also appears that drivers are able to cope with different kinds of steering wheel torque modifications without altering their driving task performance. In fact, the reaction of drivers to unexpected changes of the steering wheel torque has been investigated. In [6], Toffin et al. studied on dynamic driving simulator the behavior of drivers facing modification of the steering wheel torque rendering law before cornering maneuvers (see figure 1). The change was applied during driving on a straight road then drivers did not feel any sudden variation of the steering wheel torque.

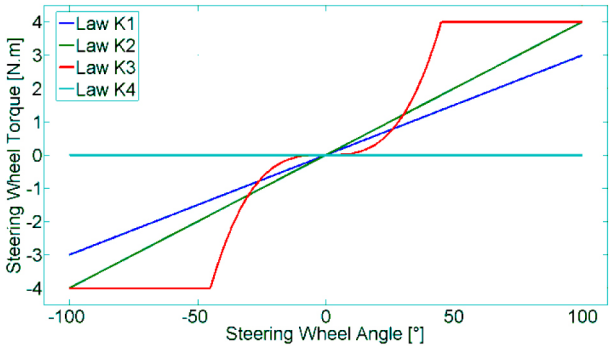



Figure 1. Different steering wheel torque rendering laws.

For K1, K2 and K3 laws, subjects were able to achieve their driving task with a high level of performance. In [7], drivers showed the capability to manage a sudden and unpredictable increase of the steering wheel torque while driving in a curve.

From a designer point of view, the appropriate and earliest tuning of such systems in the vehicle development process is a challenge to deal with. In particular, the way drivers perceive such systems activity. That's why we propose in this paper to assess the capability of driver to discriminate changes in the steering wheel torque.

## Driver Discrimination Capabilities

The human torque discrimination capabilities have been investigated in many studies. The metric commonly used to characterize the perception threshold is the Just Noticeable Difference (JND). The JND is a classical measure in psychophysics. It describes the minimum variation of a stimulus that human can detect with a probability of 50%. This probability level could be different regarding to the context.

In [8], the authors used a haptic device, which enabled torque sensing and controlling and looked like a screwdriver (Instrumented ScrewDriver). Subjects had to grasp the ISD with their dominant hand. In each trial, they experienced a reference torque and an incremented. They had to judge if the second torque was smaller or greater than the first one. Authors were able to measure a threshold of 12.7% in torque discrimination, considering a reference torque of 60mN.m.

[9] have shown that the torque discrimination capability can be influenced by the presence of an additional force on the handled device. In their experiment, subjects handled with on hand a magnetic levitation interface which was able to render different force profiles (constant, correlated or not to the torque level, random) in addition of the torque. They measured a JND varying between 15% and 43.7%, for a torque reference of 0.2N.m

The previous studies established a JND for torque discrimination with haptic devices handled with one hand. The reference torques are less than 1N.m. In the context of driving, the steering wheel is manipulated by both hands and torque levels are higher. [10] suggest drivers have a better estimation of the steering wheel force than the steering wheel torque. In fact they designed an experiment were subjects had to reproduce a reference effort on a specific steering wheel. It was made of different rims with different radius (‘small’, ‘medium’ and ‘large’) and allowed to uncouple the force-torque relationship. Then in a second experiment, they studied the discrimination capabilities for the following reference forces 5.25N, 10.5N and 21 N and the median measured difference thresholds were 16.5%, 15% and 11.5%. We also notice a 79.4% probability of subject right answers for the JND computation.

From this review, we propose to study the driver discrimination capabilities for steady state torques. Before presenting the experiment, we need to be able to control the torque-rendering device properly.

## Steering Wheel Control Law

In order to ensure a high level of torque rendering for this experiment, we investigated different control strategies. Several of them were developed for SBW vehicles. Some strategies are based on model matching theory [10-11-12]. The idea is to approximate the transfer functions of an ideal model to the real system using a control strategy. In that way, [11] implemented and compared Proportional-Derivative and LQ control laws using an electric power system as ideal model to compute the control law gains.

The RST control law, also known as polynomial control law was implemented in [14]. This control strategy has the advantage of having two degrees of freedom: a feedforward and a feedback. Despite this benefit, setting the coefficients of the polynomials is not obvious, once the resolution of the Diophantine equation guarantees the polynomials stability, but not the system ones. Nevertheless, small errors were shown in the results for an input signal up to 2 Hz.

[15] used the internal model controller concept. This technique was implemented using a dynamic model of the system and a disturbance observer. So, the influence of the driver impedance in the system was rejected. In both articles a PD controller was implemented to ensure the reference tracking. Good results

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were shown for a 5 Hz input. However, as for the RST approach, a well-known dynamic model of the system is necessary.

A PID model reference was presented in [13]. This classic technique has an ideal dynamic model as reference for the real system. The role of the control law is to reduce the difference between the model and system outputs. In [14], the PID gains were calculated iteratively in order to obtain a quasi-ideal tracking and robustness combination.

For our study, a PID tracking control was chosen as a control law for the steering wheel interface. Based on [11], this technique has presented good results on the frequency range of regular driving (0.1 to 0.6 Hz [16]) and for our test torque signals.

## Experiment

### *Experimental Setup*

The Torque Rendering Device (TRD) is based on a brushless electric motor and allows rendering up to 30N.m with a resolution of 0.08N.m and very low damping and friction. The TRD receives torque demands from the acquisition system and sends back an estimation of the real steering wheel torque based on the measure of the motor current. The communication between the TRD and the acquisition system is based on a CAN network exchanging messages at 1000Kbauds.

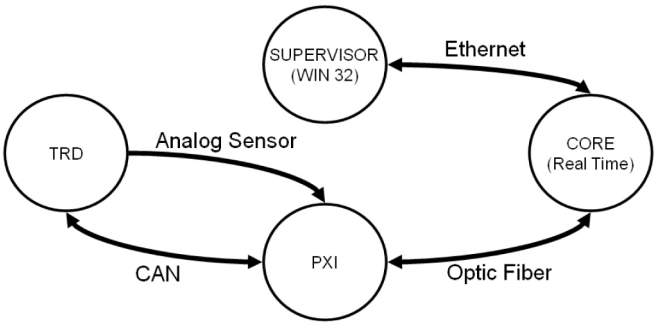



Figure 2. Acquisition system architecture

An analog sensor was mounted in the steering wheel in order to provide a measure of the torque difference between the produced torque by the TRD and the driver applied torque. An FPGA card ensured the acquisition of this sensor signal. The torque measure is produced by the deformation of constraint gauges.

The acquisition system is based on different component (see figure 2). A supervisor PC takes in charge the tuning and the control of the core state machine software. The core is running under a real time operating system. Then it allows acquiring signals, computing torque demand, and managing CAN communication and record data at a constant frequency of 1000Hz. The acquisition software is developed in Labview 8.0. The acquisition cards for analog torque sensor (FPGA cards) and CAN network are gathered in a dedicated external device: a PXI from National Instruments.

### *Protocol*

Subjects were invited to sit in front of the steering wheel interface and to set the distance to the interface as they used to in their car (see figure 3). This distance was measured which ensured to be the same during the whole experiment. The subjects were required to hold the steering wheel with both hands and to resist to efforts, maintaining the steering wheel in the center position.



Figure 3. Experimental setup

JND were determined with a two-alternative forced-choice procedure. Each trial consisted of two stimuli, i.e. two torque magnitudes, and a pause between stimuli. The reference and test stimuli were presented sequentially in a random order to subjects. Then, they were asked which one was the strongest. The profile of the stimulus curve was chosen in order to minimize the possible overshoot.

Each stimulus was presented during 1000 ms. The rise and fall of the stimulus curve took 200 ms each one. The pause between stimuli was also of 1000 ms. An example of a trial signal is shown in Figure 4, the reference torque (1N.m) is followed by the test torque (1.2N.m).

Three different reference magnitudes were chosen to be analyzed: 1, 3, and 5 Nm. These magnitudes were chosen from the range of steering torque feedback in a regular driving situation. The paradigm was divided in one session for each torque reference.

In each session, the torque reference was presented in two directions: right (“clockwise” condition) and left (“counter-clockwise”). For each direction, 16 increments were chosen: 8 positives (“higher” condition) and 8 negatives (“lower”) based on the TRD resolution and JND obtained from the literature. We presented each trial three times.

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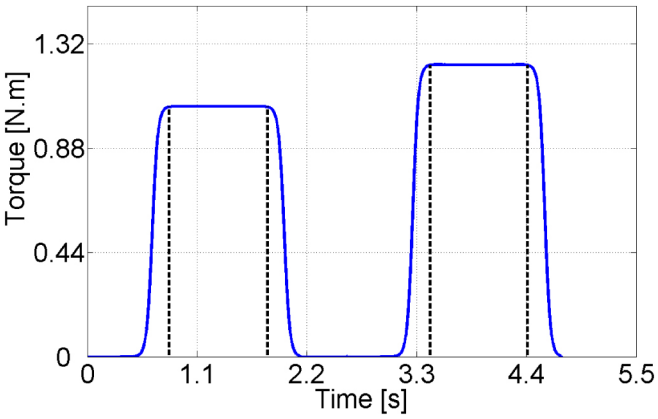



Figure 4. Example of trial stimuli

Therefore, 307 trials in total: 3(repetitions) × 3(torque references) × 2(directions) × (16(increments) + 1(reference torque)). There were 5 min break every 51 trials for subject rest and whenever the he desired. Then, the equation 1 describes the computation of the JND where *Tr* is the reference torque and *dT* is the torque variation.

$$JND = 100 \cdot \left| \frac{(Tr + dT) - Tr}{Tr} \right| \quad (1)$$

As in [10], we used a JND based on a probability of 79.4% of subject right answers.

## Results

### *Subjects*

Eighteen subjects, aged between 22 and 56 years, participated in the experiment. All of them had at least a previously 4 years driving experience. The presentation order for the reference torques was balanced across subjects, aiming an equal number of subjects for each orders (see table t1).

Table 1. Presentation order

| Subject | Stimuli presentation order in N.m | | |
|---|---|---|---|
| 1 | 1 | 5 | 3 |
| 2 | 5 | 3 | 1 |
| 3 | 5 | 1 | 3 |
| 4 | 3 | 5 | 1 |
| 5 | 3 | 1 | 5 |
| 6 | 1 | 3 | 5 |
| 7 | 1 | 5 | 3 |
| 8 | 5 | 3 | 1 |
| 9 | 5 | 1 | 3 |
| 10 | 3 | 5 | 1 |
| 11 | 3 | 1 | 5 |
| 12 | 1 | 3 | 5 |
| 13 | 1 | 5 | 3 |
| 14 | 5 | 3 | 1 |
| 15 | 5 | 1 | 3 |
| 16 | 3 | 5 | 1 |
| 17 | 3 | 1 | 5 |
| 18 | 1 | 5 | 3 |

### *Torque Reference 1N.m*

For the 1N.m torque reference, the test torque range was from 0.48N.m to 1.6N.m. The figure 5 shows the perception curve built from subjects' response. For all the perception curves presented here, both directions (clockwise and counter clockwise) are displayed and the red dash line represents a 79.4% probability of right answers.

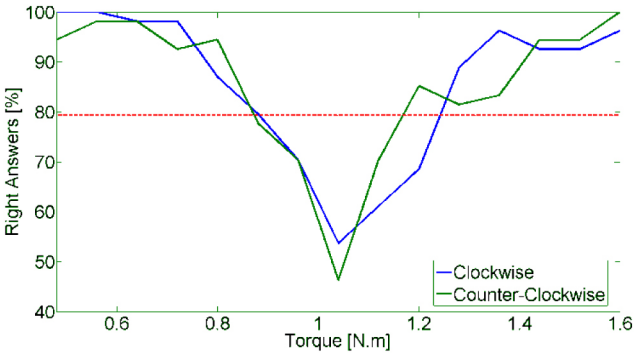



Figure 5. Perception curve for torque reference of 1N.m

### *Torque Reference 3N.m*

The test torque range for the 3N.m reference torque condition was from 2.24N.m to 3.84N.m. As for the 1.N.m condition, the perception curce is displayed on the figure 6.

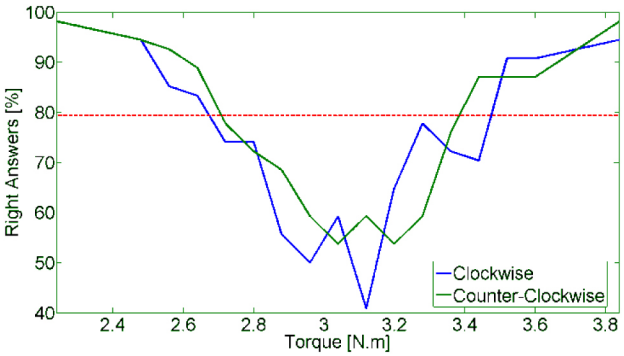



Figure 6. Perception curve for torque reference of 3N.m

### *Torque Reference 5N.m*

Then the figure 7 shows the perception curves for the last reference torque condition. The range of tested values was [4.08N.m 6.0N.m].

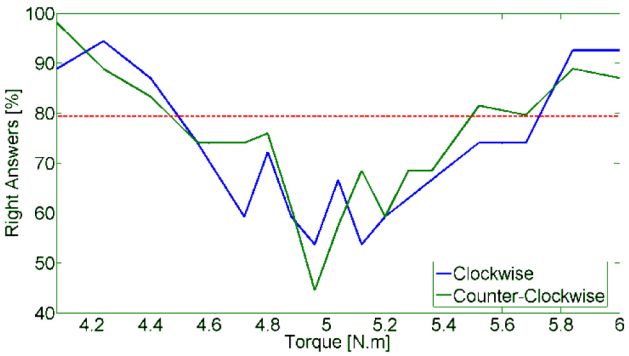



Figure 7. Perception curve for torque reference of 5N.m

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## Discussion

During the 1N.m torque reference condition, subjects had the best ratio of right answers than for the other torque reference conditions, as reported in table 1.

Table 1. Percentage of right answers

| Reference | Clockwise | Counter clockwise |
|---|---|---|
| 1N.m | 82.3% | 82% |
| 3N.m | 74.1% | 71.6% |
| 5N.m | 70.7% | 68.5% |

Subjects also said the 5N.m torque reference condition was the most difficult to achieve due to the amount of effort to provide.

The table 2 shows the mean JND over whole subjects for the three torque references and the counter clockwise steering wheel rotation direction. We notice that the highest JND was measured for the lowest reference torque condition. Moreover, it appears that the JND decreases over the increase of reference torque level, which tendency is observed in [10].

Table 2. JND results, counter clockwise condition

| Reference | Clockwise | Counter clockwise |
|---|---|---|
| 1N.m | 14,8% | 18.0% |
| 3N.m | 11.2% | 13.3% |
| 5N.m | 10.2% | 12.7% |

We also notice a difference between clockwise and counter clockwise conditions. The fact that during the experiment, subjects systematically experienced the clockwise direction and after the counter clockwise could be an explanation of this phenomenon. It also appears that subjects had more difficulties to discriminate torque level when the test torque was lower than the reference torque (“lower” condition). In addition with subjects' fatigue, this could explain the high levels of JND in the counter clockwise condition.

Besides, we have to remind some specification of human limbs. In fact, muscle fibers can only generate contracting force on joints. The Central Nervous System recruits muscles in couple, an agonist and its antagonist, in order to produce a force difference on joints allowing two ways movements. Then more or less muscle fibers can be used to produce a same torque level. This phenomenon allows the modulation of limbs impedance at each joint to improve robustness and accuracy [17]. This means it could interest to measure the level of muscles contraction in such experiments to assess an eventual effect on the discrimination capabilities. Then, it would be also interesting to assess how the muscle fatigue factor could have effect on the JND values.

Then, the measured JND was established for a non-driving situation. Subjects were able to fully concentrate on the experiment without handling a vehicle trajectory. We can expect that performing a task as complex as driving could have an effect on driver discrimination capabilities.

## Applications

*Driving simulator.* As explained in [6], steering wheel torque feedback is an important cue for the driving task. Particularly for uncoupled steering vehicle and driving simulators, high quality torque rendering is essential. This requires accurate vehicle steering mathematical model, high performance torque rendering device and control strategy. In that context, our results can help in the development of control laws and the assessment of their performances such as acceptable steady state error.

*Driver adaptation capabilities and safety studies.* As seen previously, different studies suggest drivers are able to cope with different torque variation on the steering wheel. In the context of a sudden and detectable torque level increase or decrease, drivers are able to adapt to such events. This highlights the adaptive processes (impedance modulation and internal representation update [18]) capabilities. In the context of undetectable changes, the question about the performance of these strategies arises. This study gives cues on the on driver discrimination capabilities assessment. It would be necessary to extend this experiment for driving task condition and sudden torque changes.

## Conclusion

In this paper, we proposed an experiment to assess the driver discrimination capabilities to steering wheel torque variations, in non-driving condition. Three levels of torque were studied (1N.m, 3N.m and 5N.m) corresponding to the range of torque that drivers can encounter on steering assisted automotives. We measured in clockwise direction (respectively counterclockwise) a JND varying from 14.8% (respectively 18%) to 10.2% (respectively 12.7%). As the probability of right answers decreased with the increase of the torque level, we also observed a decreasing JND value. Then, subjects reported their difficulty to perform the required task with the highest torque level.

This study highlighted subjects capability to compare only steady state torques in non-driving condition. Future work will focus on both driver capabilities to detect transitory torques and during driving task.

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## Definitions/Abbreviations

**ADAS** - Advanced Driving Assistance System

**CNS** - Central Nervous System

**ISD** - Instrumented ScrewDriver

**JND** - Just Noticeable Difference

**LQ** - Linear Quadatric

**PD** - Proportional Derivative

**PID** - Proportional Integrator Derivative

**SBW** - Steer-By-Wire

**TRD** - Torque Rendering Device

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